

Performance Testing

Date	4 july 2026
Team ID	SWTID-2026-5594
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Web Browser (Chrome), Flask Development Server
Type of Testing	Load Testing (Basic Functional Performance Test)
Target Module / API	/predict endpoint
Test Environment	Local Machine (Windows 11, VS Code, Python 3.12, Flask)
Test Date	05 July 2026

Step 2: Test Scenarios

S.No	Test Scenario	No. of Virtual Users	Duration	Expected Outcome	
1	Single user prediction	1	30 sec	Prediction displayed successfully	
2	Multiple prediction requests	10	60 sec	System remains responsive	
3	High income applicant	1	20 sec	Prediction generated successfully	
4	Low income applicant	1	20 sec	Prediction generated successfully	

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Average Response Time	< 2 sec	0.35 sec	Pass	Fast prediction
2	Maximum Response Time	< 5 sec	1.10 sec	Pass	Stable
3	Throughput	-	20 Requests/sec	Pass	Suitable for demo
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	32%	Pass	Normal usage
6	Memory Utilization	< 80%	41%	Pass	Stable memory usage

Step 4: Observations & Analysis

- The Flask web application responded quickly to user requests during local testing.
- The Random Forest model generated predictions within a short response time.
- No application crashes or critical runtime errors were observed during functional testing.
- The system remained stable while processing repeated prediction requests.
- The application is suitable for small-scale deployment and demonstration purposes.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

